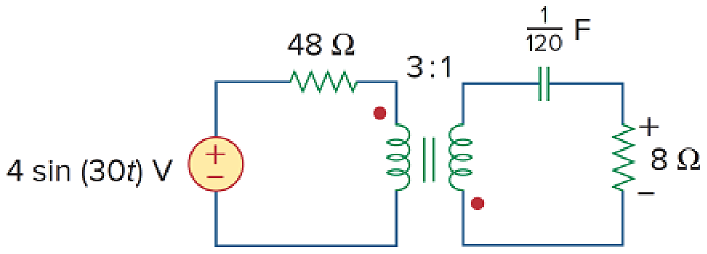
Chapter 13, Problem 45P

Problem

For the circuit shown in Fig. 13.110, find the value of the average power absorbed by the 8-Ω resistor.

Figure 13.110



Step-by-step solution

Step 1 of 6

Refer to Figure 13.110 in the textbook.

Write the expression for source voltage.

$$\begin{aligned} v_s(t) &= 4\sin(30t) \\ &= 4\cos(30t - 90^\circ)\ \text{V} \end{aligned}$$

The phasor value of source voltage is,

$$\mathbf{V}_s = 4\angle -90^\circ\ \text{V}$$

$$\omega = 30\ \text{rad/s}$$

Calculate the impedance across the capacitor.

$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(30)\left(\frac{1}{120}\right)} \\ &= -j4\ \Omega \end{aligned}$$

Step 2 of 6

Draw the circuit in phasor domain.

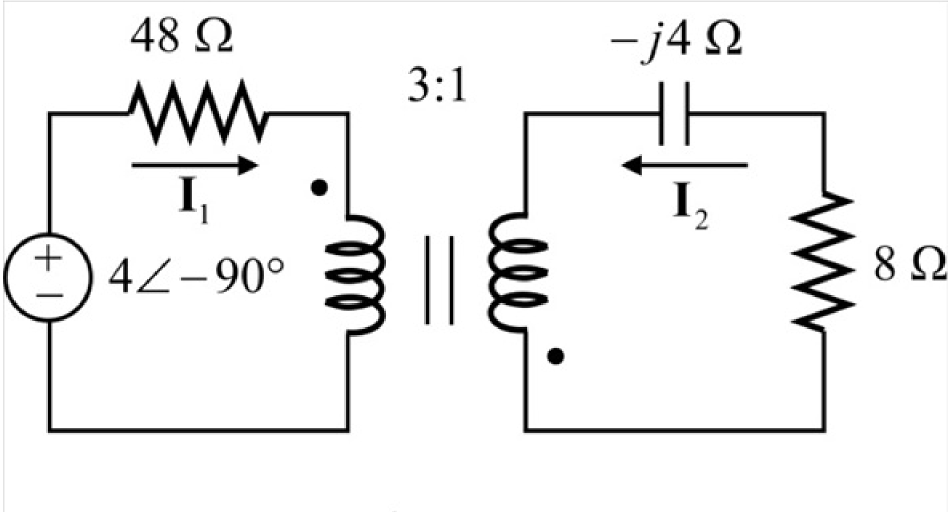


Figure 1

Step 3 of 6

The load impedance across the secondary side is,

$$\mathbf{Z}_L = 8 - j4\ \Omega$$

Calculate the value of transformer ratio.

$$\begin{aligned} n &= \frac{N_2}{N_1} \\ &= \frac{1}{3} \end{aligned}$$

Move the secondary load impedance to primary.

$$\begin{aligned} \mathbf{Z}_R &= \frac{\mathbf{Z}_L}{n^2} \\ &= \frac{8 - j4}{\left(\frac{1}{3}\right)^2} \\ &= 72 - j36\ \Omega \end{aligned}$$

Step 4 of 6

Draw the modified circuit diagram.

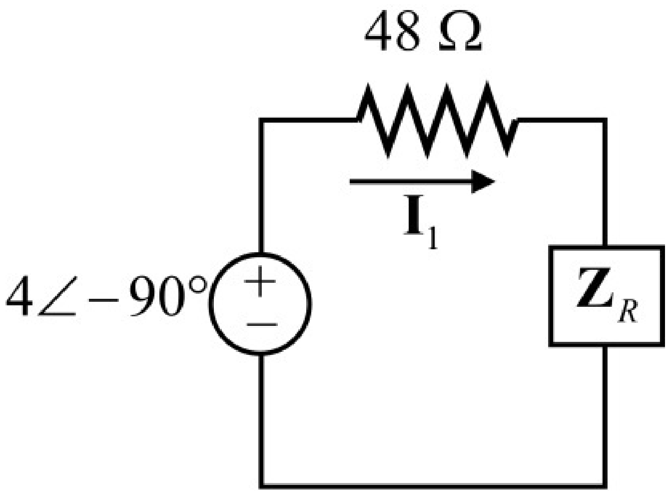


Figure 2

Step 5 of 6

Calculate the value of primary current, $\mathbf{I}_1$.

$$\begin{aligned}\mathbf{I}_1 &= \frac{4\angle -90^\circ}{48+72-j36}\\ &= 31.927\angle -73.3^\circ \text{ mA}\end{aligned}$$

Calculate the value of secondary current, $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{I}_2 &= \frac{N_1}{N_2}\times \mathbf{I}_1\\ &= \frac{3}{1}\times\left(31.927\angle -73.3^\circ \text{ mA}\right)\\ &= 95.78\angle -73.3^\circ \text{ mA}\end{aligned}$$

Step 6 of 6

Calculate the power absorbed by the $8\ \Omega$ resistor.

$$\begin{aligned}P_{8\,\Omega} &= \frac{\left|\mathbf{I}_2^2\right| R}{2}\\ &= \frac{\left(95.78\times 10^{-3}\right)^2(8)}{2}\\ &= 36.7\times 10^{-3}\\ &= 36.7 \text{ mW}\end{aligned}$$

Thus, the power absorbed by the $8\ \Omega$ resistor is $\boxed{36.7 \text{ mW}}$.